

런던 주택 가격 예측

데이터마이닝 방법 및 실습 프로젝트 제안서
[팀 4] 박태현, 오종현, 이예찬, 박창현

Motivation

한국경제 ☒ 구독중

PICK ⓘ

"요즘 쿠팡으로 장 보죠"...집값 1억2000만원 차이나는 이유 [오세성의 헌집만세]

입력 2024.03.30. 오전 7:02 · 수정 2024.03.30. 오후 7:10 기사원문



오세성 기자



85



180



가가



오세성의 헌집만세(5)

"구축은 무조건 입지 좋을 줄 알았는데"
도로 하나 차이로 엇갈리는 새벽 배송... 집값도 영향

"로켓프레시 제공해 달라" 아파트 주민들 단체 민원 넣기도
스타벅스 '스세권', 다이소 '다세권'도 인기

Background



수요와 공급



인구 구성

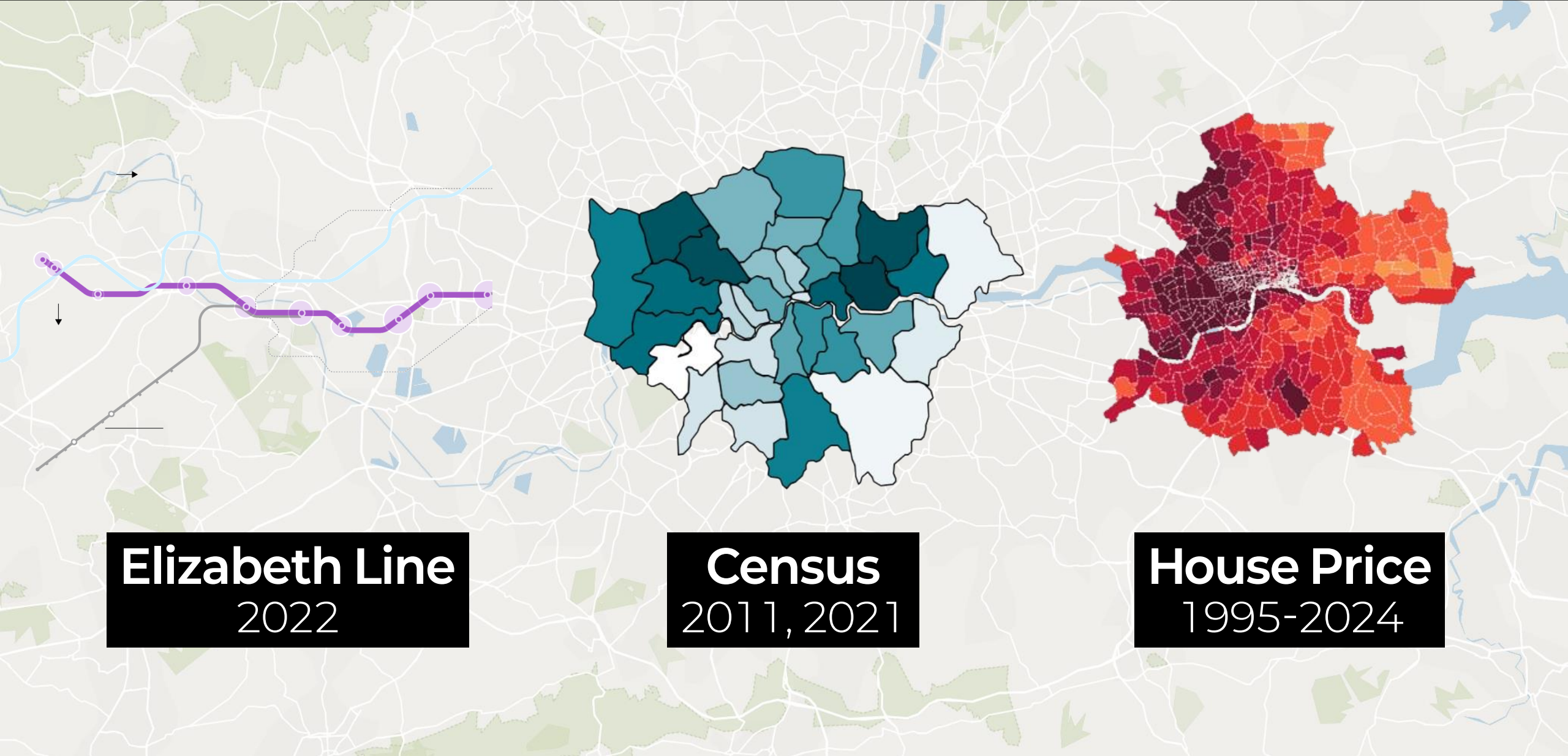


교통 인프라

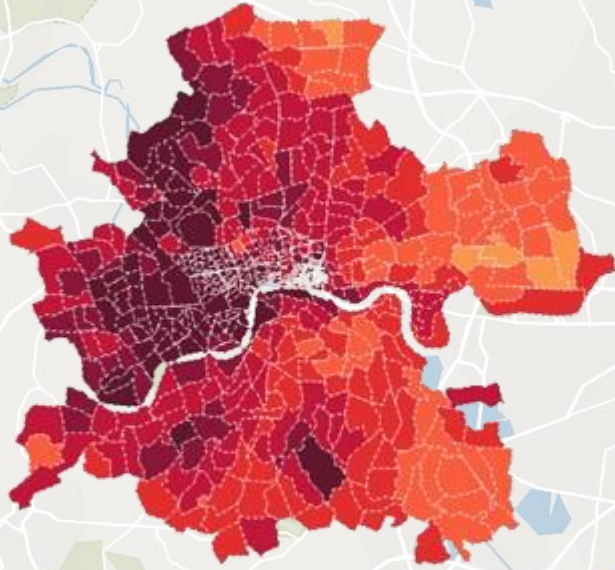


주택 가격

Purpose



Data 1: London House Price Data



N = 137,760
1995-2024

Full Address

Postcode

Latitude, Longitude

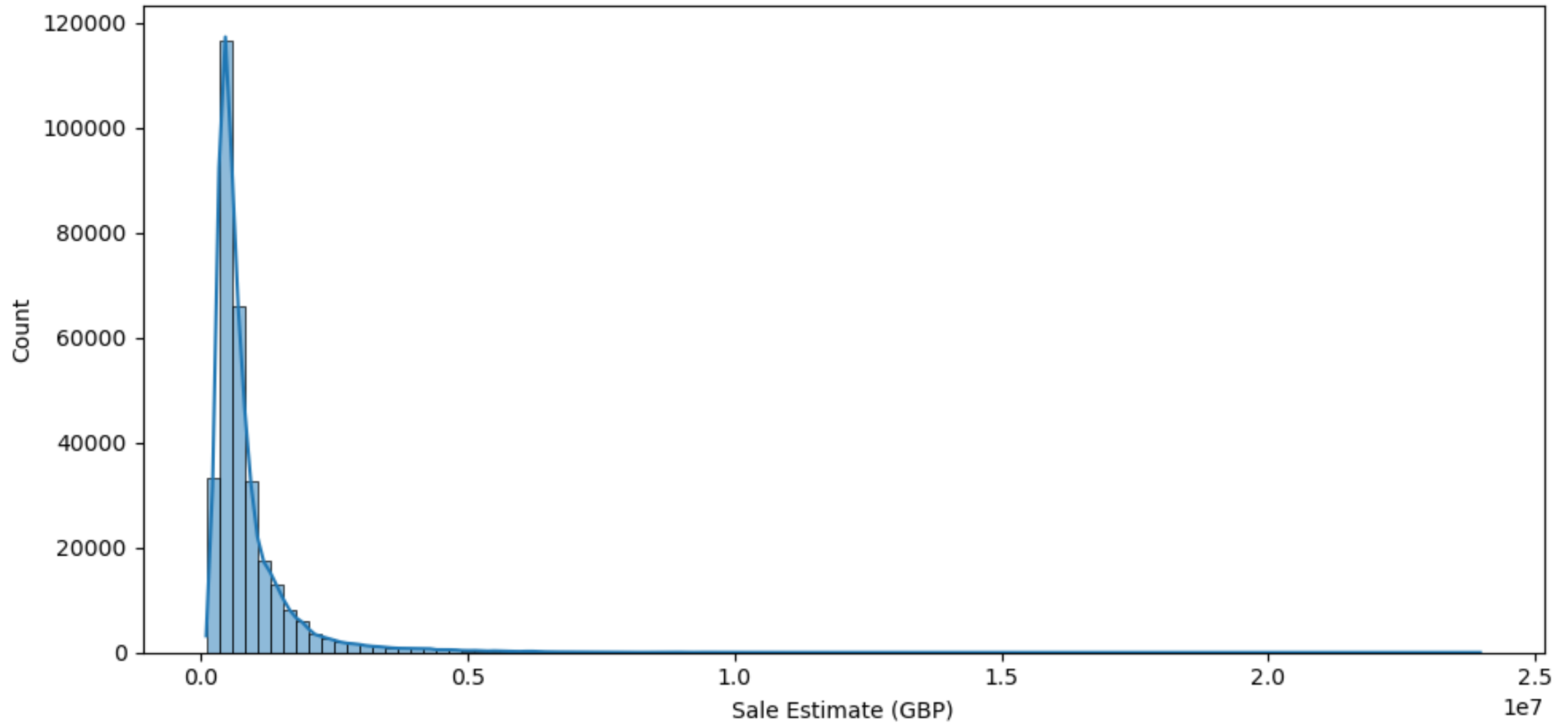
Number of Rooms, Floor Area
Bathrooms, Bedrooms, Living rooms ...

Price (Current, Lower, Upper)
Rent, Sale

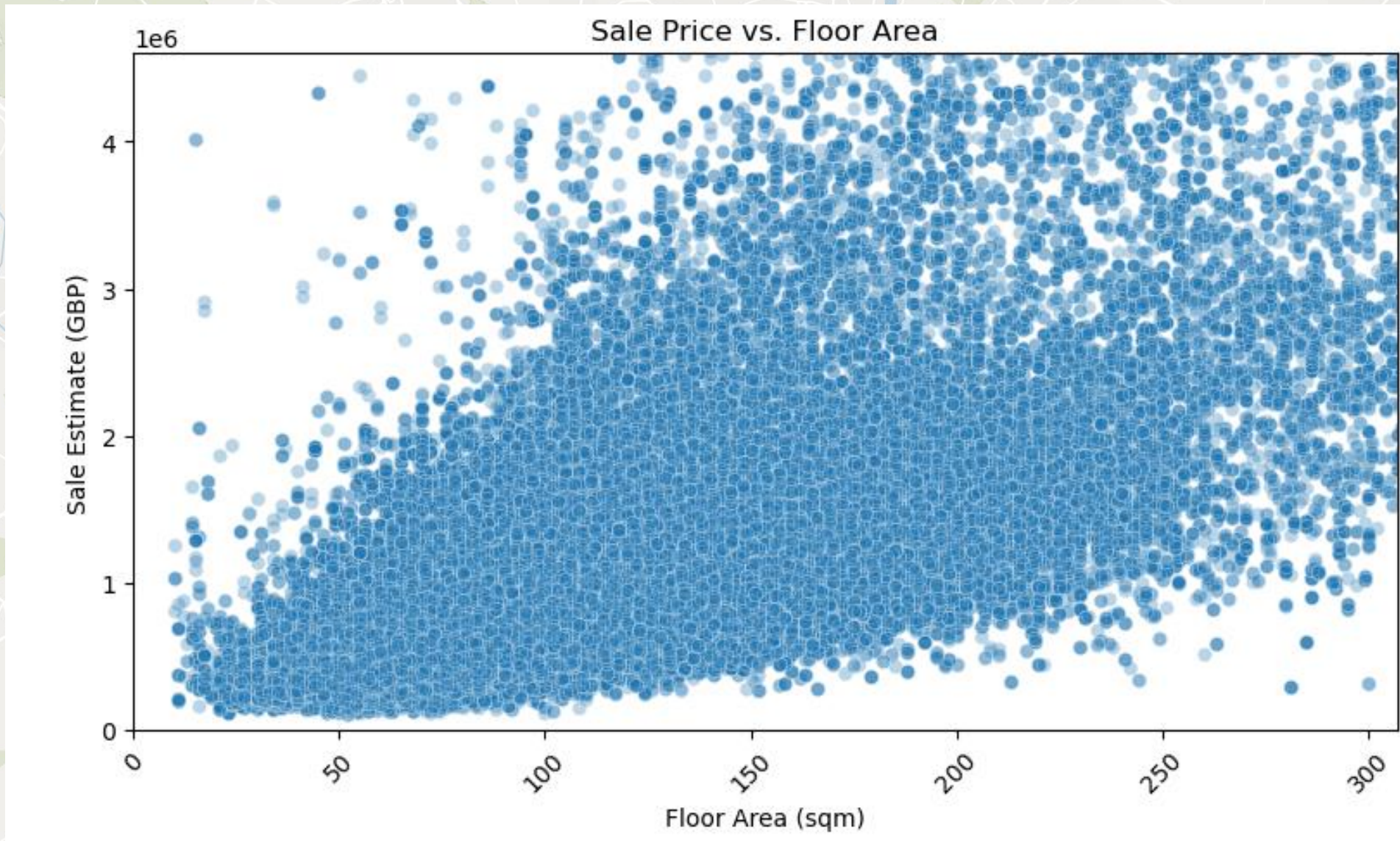
Property Type

EDA

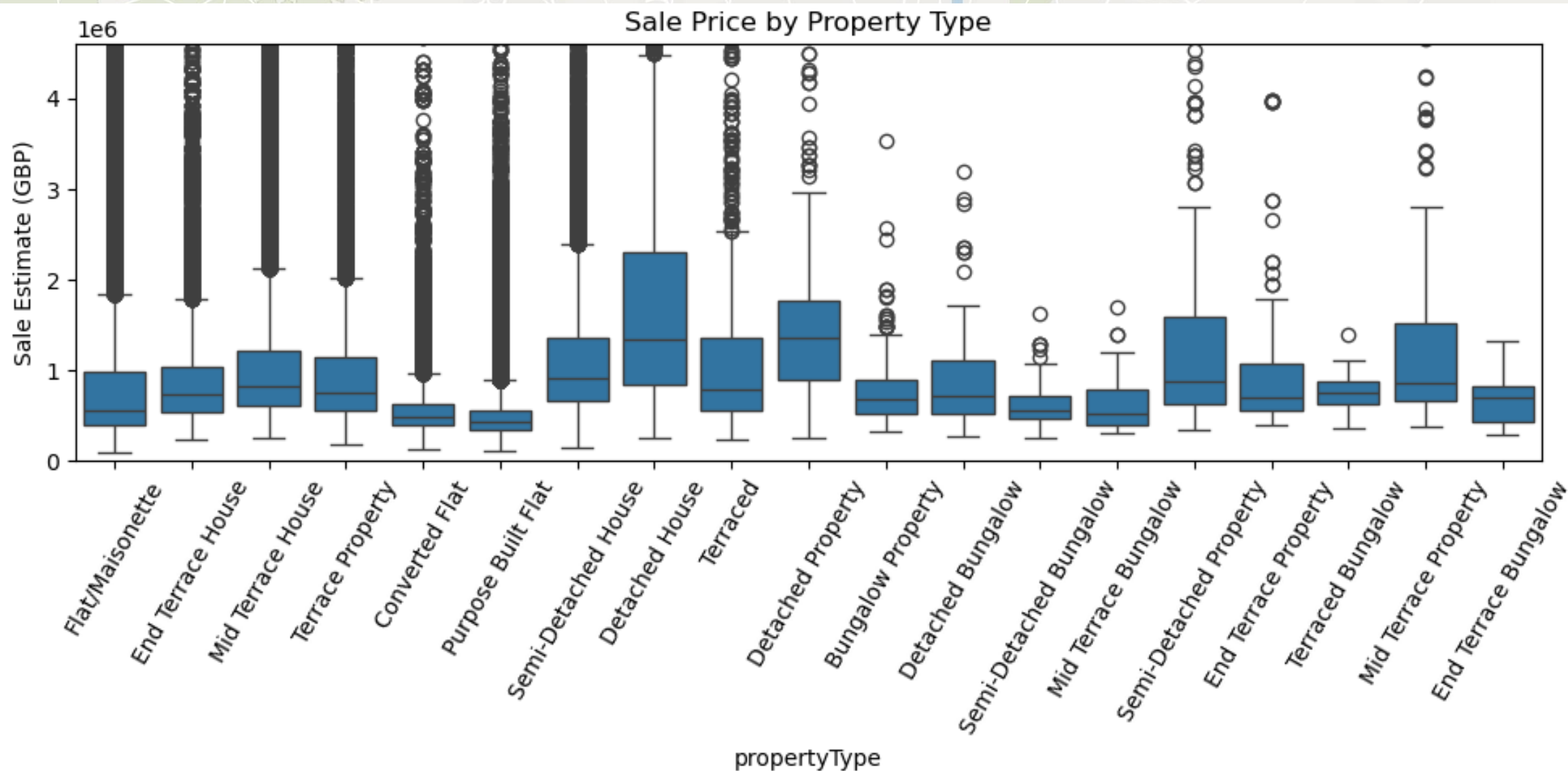
Distribution of Current Sale Price



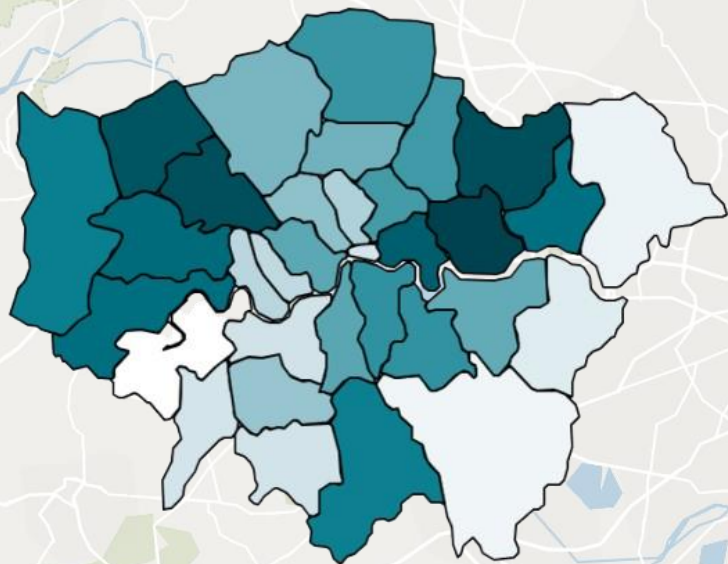
EDA



EDA



Data 2: Census Data



N = 681
2011, 2021

Household Language

- How many people in house can speak English?
- How well do people speak English?

Religion

- Christian, Muslim, Buddhist, Hindu ...

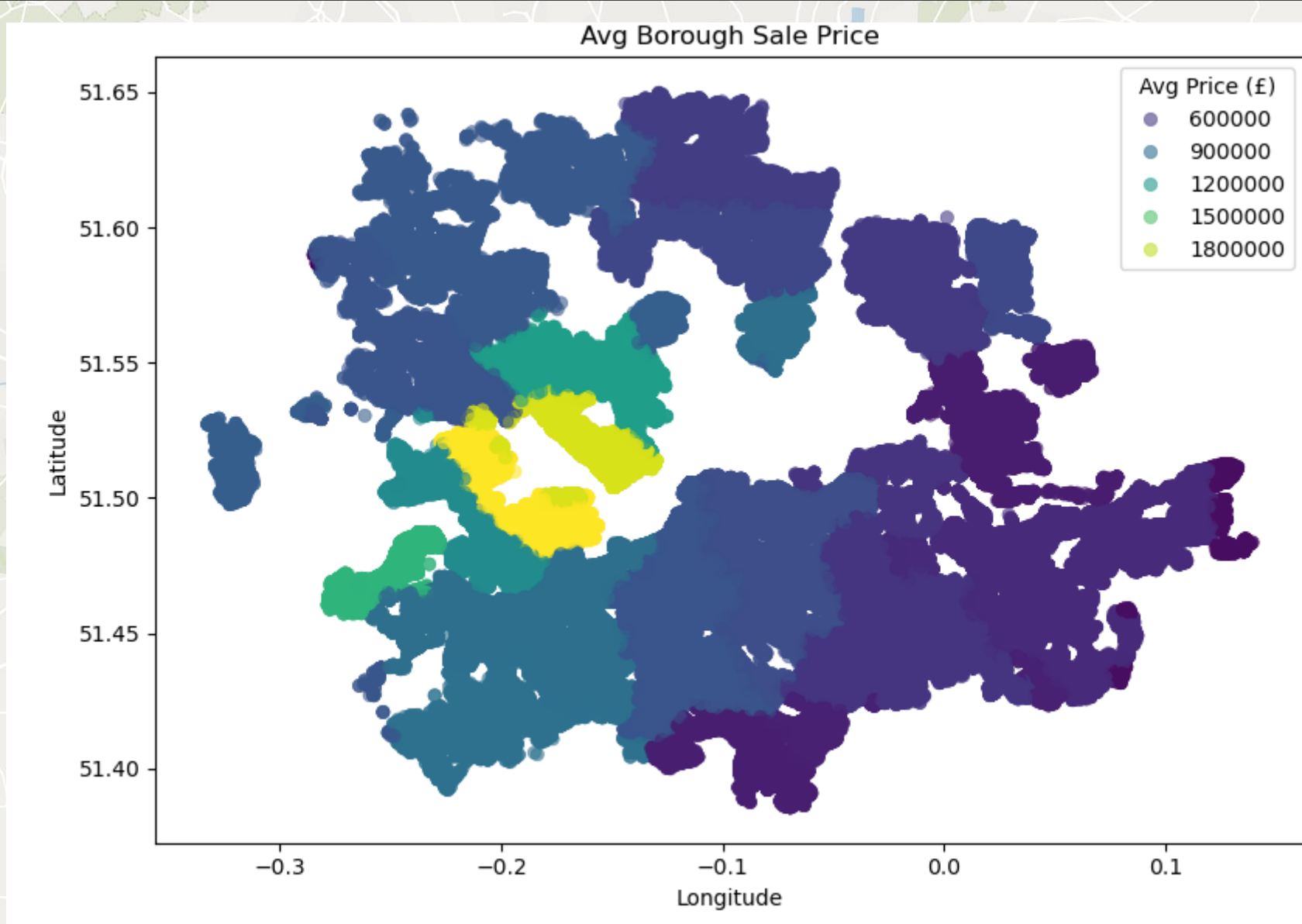
Ethnic Group

- White British, White Irish, Mixed, Asian, Black ...

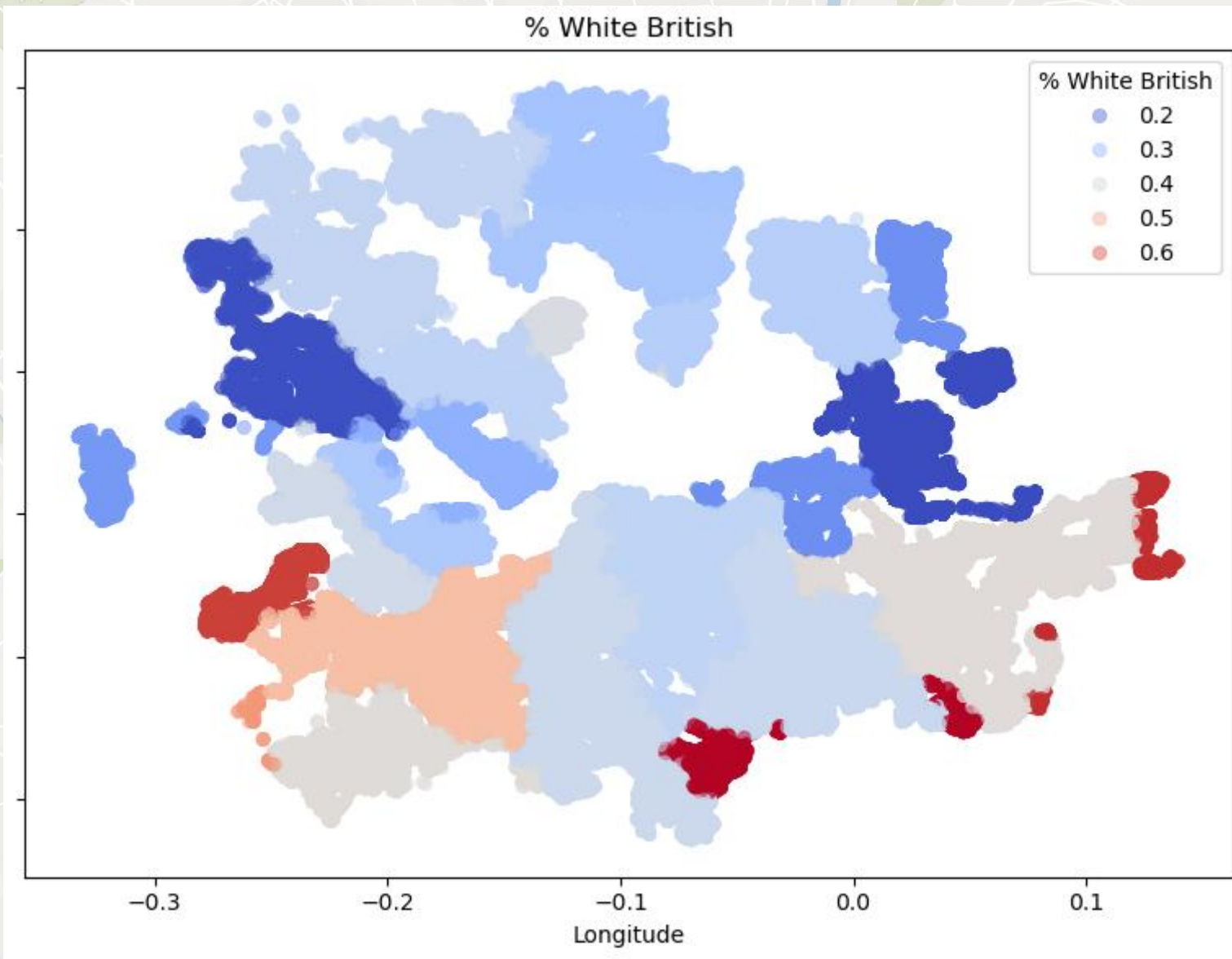
EDA



EDA

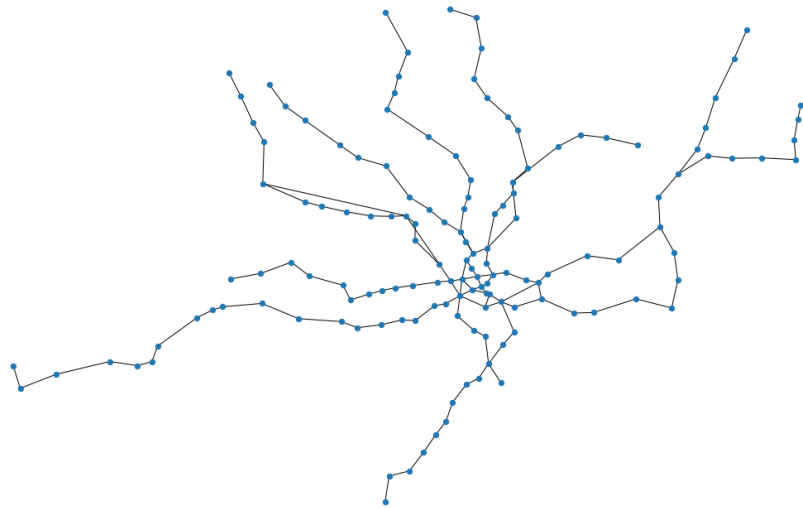


EDA



Data 3: London Underground Data

Transport Systems & Logistics Laboratory | **Imperial College London**



Location Information
Latitude, Longitude, Line

Transport for London
London Underground

1.2. Train Predictions Service – Detailed

This service will return detailed train prediction information for a nominated station. A line and station must be nominated for this service. The detailed train prediction service returns similar data to the summary service with the addition of more detailed information. The extra data returned includes the time from station represented in seconds, the track identifier of the current location of the train and the departure time of the train.

1.3. Station Status

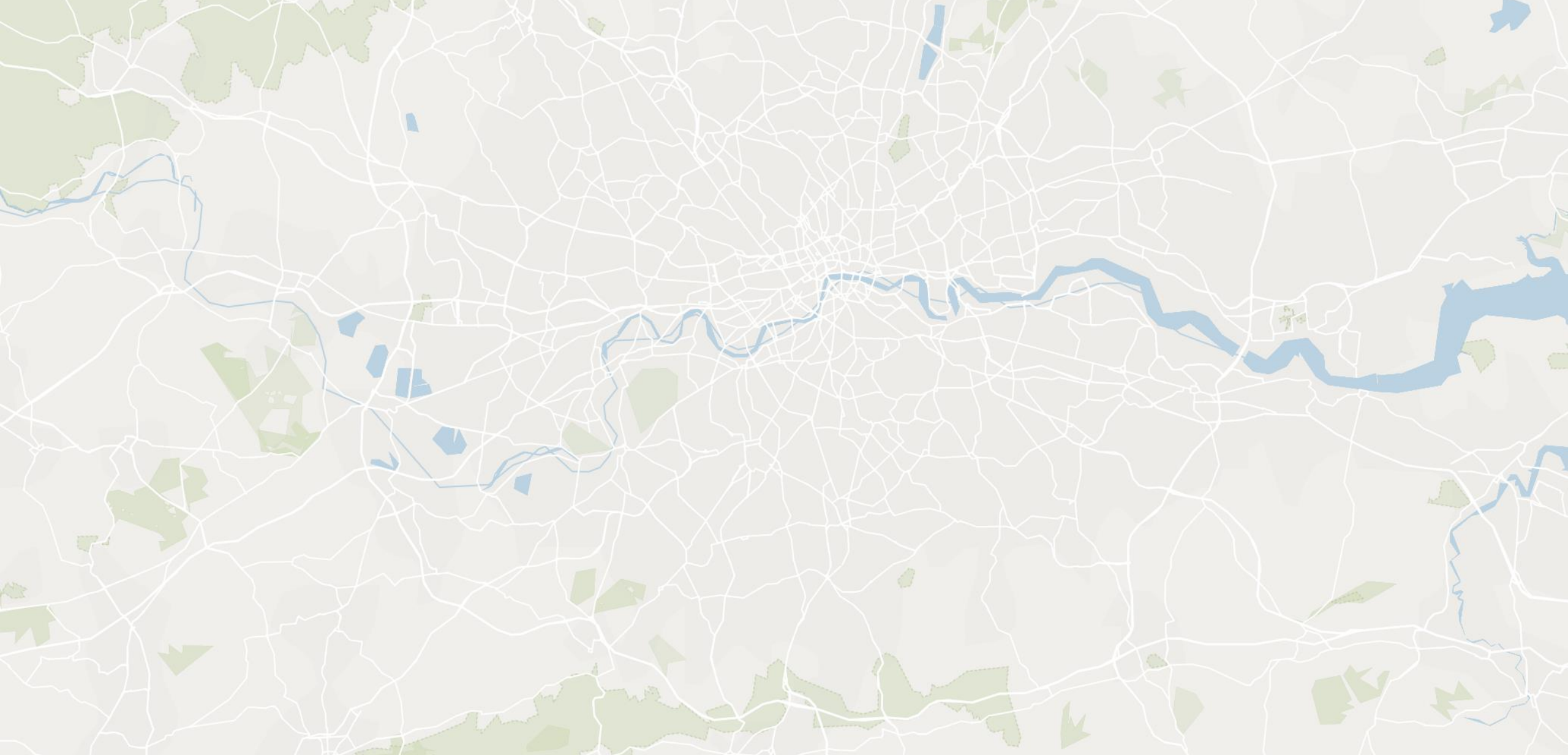
This service will return the status of all stations on the network indicating station closures, faulty lifts etc. The service can optionally be run to only return stations where the status is not normal. The service will return the station names and the current status of the station indicating what the problem is in the form of a code and a description.

1.4. Line Status

This service will return the status of all lines on the network indicating any delays, disruptions or suspensions on the lines. The service can optionally be run to return only lines where there are abnormal operations. The service will return the line names and the current status of the line indicating what the problem is in the form of a code and a description.

Train Predictions Service
Interval, Time, Direction

Analysis Plan



References

[Motivation]

<https://n.news.naver.com/mnews/article/015/0004966609?sid=101>

[Data 1]

<https://www.kaggle.com/datasets/jakewright/house-price-data/data>

[Data 2]

<https://data.london.gov.uk/dataset/2021-census-wards-ethnicity-language-identity-religion>

[Data 3]

https://transport-systems.imperial.ac.uk/tf/notebooks/n05_studying_the_london_underground.html

<https://tfl.gov.uk/info-for/open-data-users/our-open-data>